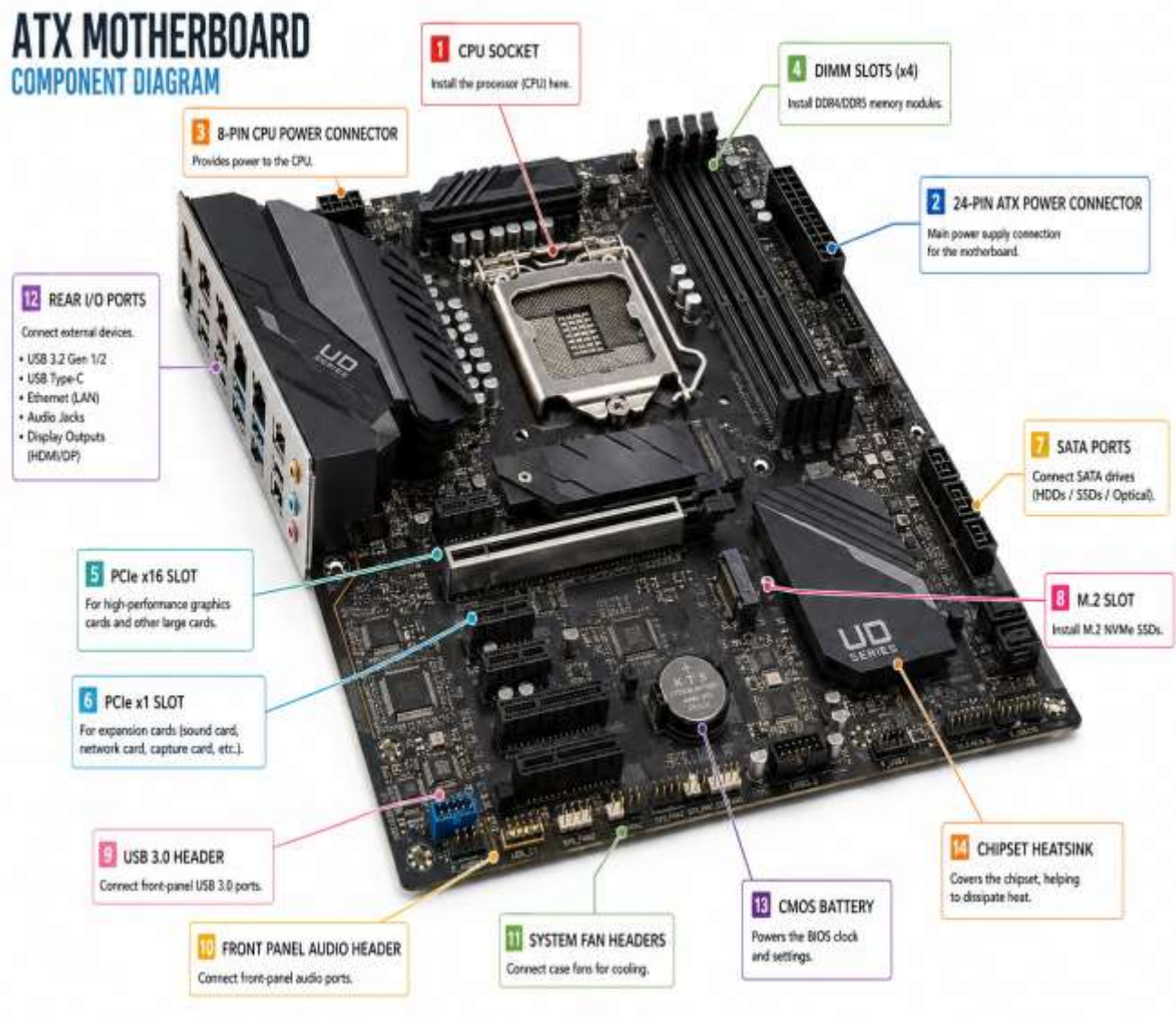


Assignment-2

I.

Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2.

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

CPU Socket Type	Compatible CPU Model	Company
LGA1151	Intel Core i7-9700K	Intel
AM4	AMD Ryzen 5 5600X	AMD
LGA1200	Intel Core i5-10400	Intel

1. LGA1151

- **Company:** Intel
- **Compatible CPU:** Intel Core i7-9700K
- **Used in:** 8th and 9th Generation Intel Core processors.

2. AM4

- **Company:** AMD
- **Compatible CPU:** AMD Ryzen 5 5600X
- **Used in:** AMD Ryzen 1000, 2000, 3000, and 5000 series processors.

3. LGA1200

- **Company:** Intel
- **Compatible CPU:** Intel Core i5-10400
- **Used in:** 10th and 11th Generation Intel Core processors.

3.

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

1. CPU (Central Processing Unit)

The **CPU** is called the "**brain of the computer.**" It processes instructions, performs calculations, and controls all the tasks running on the computer.

Main Functions:

- Executes programs and applications.
 - Performs calculations and logical operations.
 - Controls the overall operation of the computer.
 - Communicates with RAM, storage, and other hardware.
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2. Chipset (Motherboard Chipset)

The **chipset** is like a **traffic manager** on the motherboard. It controls how data moves between the CPU and other components such as RAM, SSD/HDD, USB ports, graphics card, and network devices.

Main Functions:

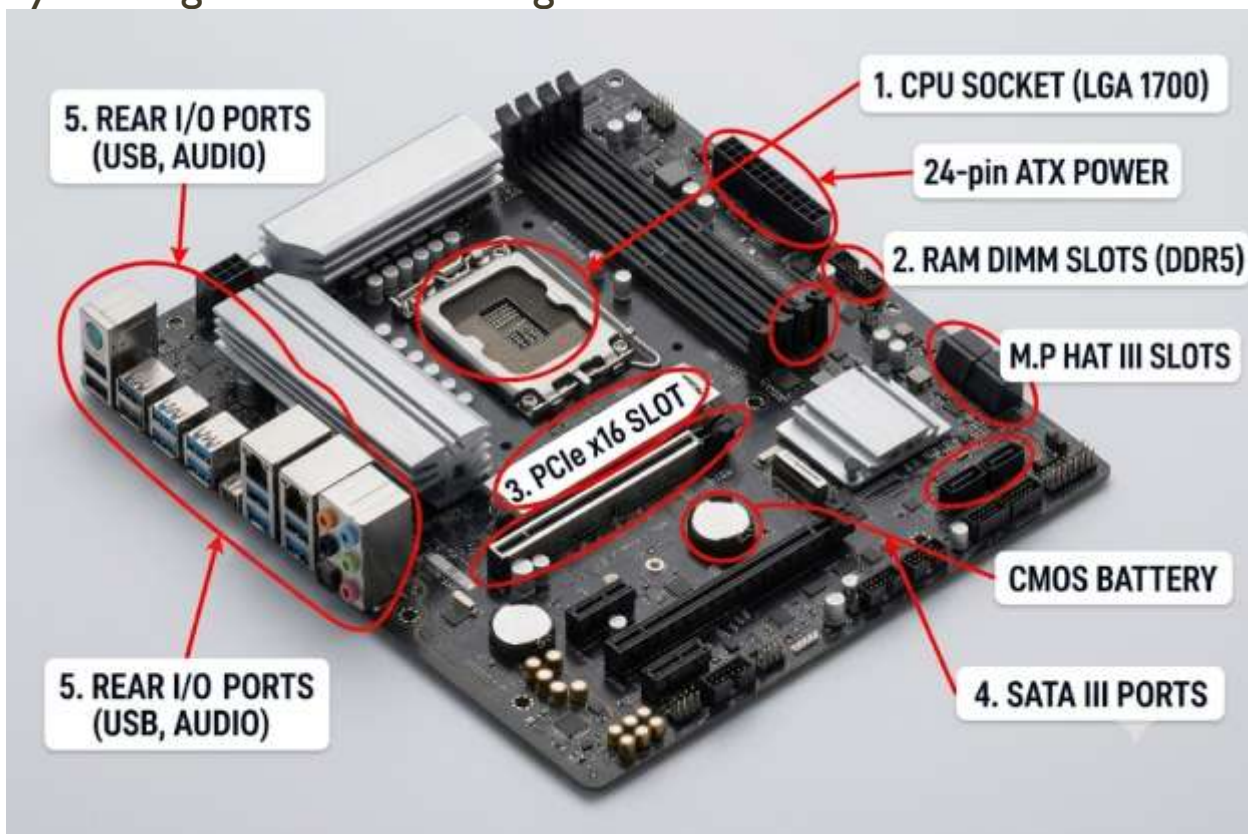
- Connects the CPU to other hardware.
 - Manages communication between different devices.
 - Controls USB ports, SATA ports, PCIe slots, and storage devices.
-

➤ How to use Music app Spotify

When you open **Spotify**, the CPU receives the command to open the app. The chipset retrieves the program from the SSD, loads it into RAM, and allows the CPU to process the data. Finally, the CPU runs Spotify while the chipset helps transfer audio and other data to the correct hardware.

4.

Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.



1. **CPU Socket:** The primary slot where the main processor (CPU) is installed to carry out system operations.
 2. **RAM (DIMM) Slots:** High-speed memory slots used to hold the system's short-term working memory (DDR RAM)
 3. **PCIe x16 Slot:** The primary high-speed expansion slot used to connect modern dedicated graphics cards (GPUs).
 4. **SATA Ports:** Dedicated storage interfaces used to connect hard drives (HDDs) and solid-state drives (SSDs).
 5. **24-pin ATX Power Connector:** The main power input socket that receives main electricity from the power supply unit (PSU).
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